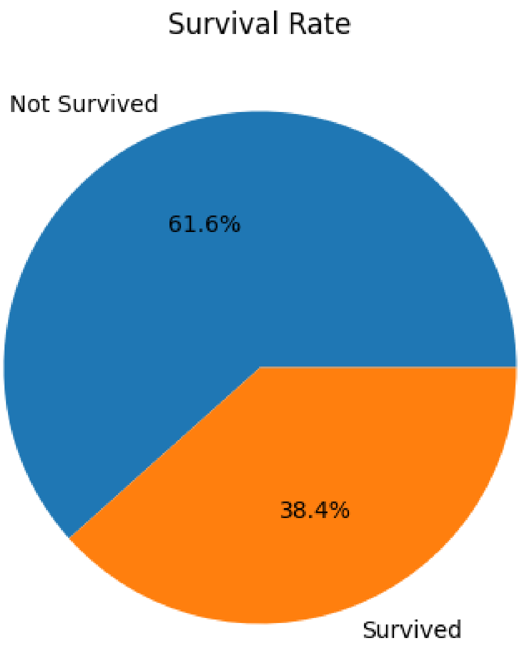
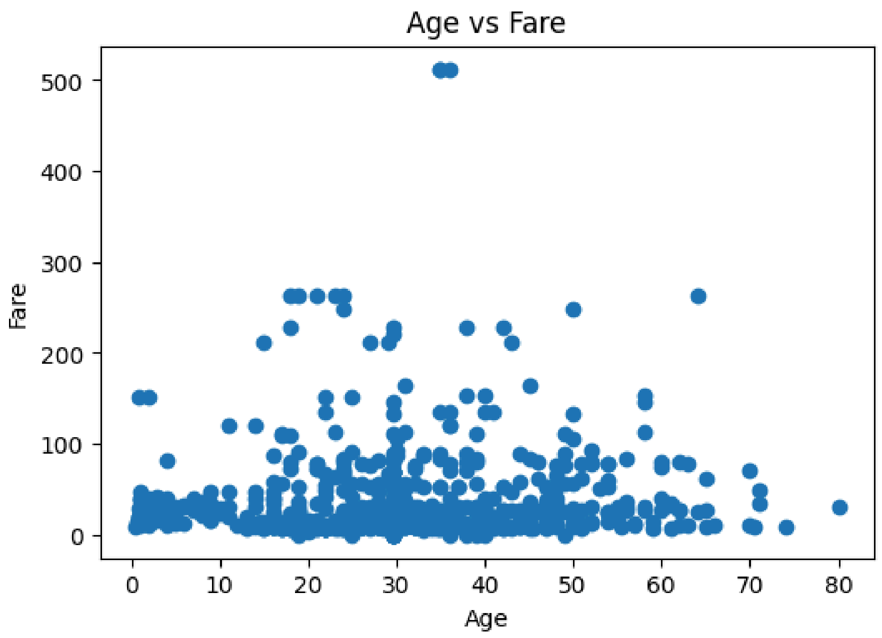
```python
import pandas as pd

# Titanic Dataset Load
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)

print("First 5 Rows")
print(df.head())

print("\nDataset Info")
print(df.info())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Rows")
print(df.duplicated().sum())

# Missing values fill
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Cabin column remove
df = df.drop(columns=["Cabin"])

# Remove duplicates
df = df.drop_duplicates()

# Save cleaned dataset
df.to_csv("cleaned_titanic.csv", index=False)

print("\nData Cleaning Completed Successfully!")
```

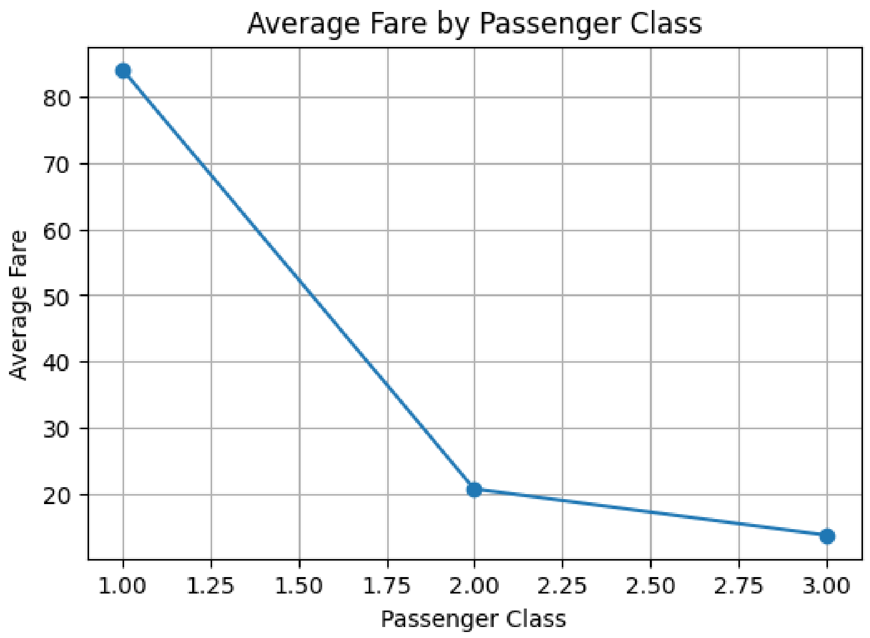
```
First 5 Rows
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S

Dataset Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
```

```
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

Missing Values
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("cleaned_titanic.csv")

# Pie Chart
plt.figure(figsize=(5,5))
df["Survived"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%",
    labels=["Not Survived","Survived"]
)
plt.title("Survival Rate")
plt.ylabel("")
plt.show()

# Scatter Plot
plt.figure(figsize=(6,4))
plt.scatter(df["Age"], df["Fare"])
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()

# Line Chart
plt.figure(figsize=(6,4))
df.groupby("Pclass")["Fare"].mean().plot(kind="line", marker="o")
plt.title("Average Fare by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Average Fare")
plt.grid(True)
plt.show()

print("Task 3 Completed Successfully!")
```








```
Task 3 Completed Successfully!
```

```
import pandas as pd
import matplotlib.pyplot as plt

# Load cleaned dataset
df = pd.read_csv("cleaned_titanic.csv")

# Summary
print("Summary Statistics")
print(df.describe())

# Survival Count
print("\nSurvival Count")
print(df["Survived"].value_counts())

# Passenger Class Count
print("\nPassenger Class")
print(df["Pclass"].value_counts())

# Average Age
print("\nAverage Age:", df["Age"].mean())

# Bar Chart
df["Pclass"].value_counts().plot(kind="bar", title="Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.show()

# Histogram
df["Age"].plot(kind="hist", bins=20, title="Age Distribution")
plt.xlabel("Age")
plt.show()
```

```
Summary Statistics
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  891.000000  891.000000   
mean    446.000000    0.383838    2.308642   29.699118    0.523008   
std     257.353842    0.486592    0.836071   13.002015    1.102743   
min       1.000000    0.000000    1.000000    0.420000    0.000000   
25%     223.500000    0.000000    2.000000   22.000000    0.000000   
50%     446.000000    0.000000    3.000000   29.699118    0.000000   
75%     668.500000    1.000000    3.000000   35.000000    1.000000   
max     891.000000    1.000000    3.000000   80.000000    8.000000   

            Parch        Fare  
count  891.000000  891.000000  
mean     0.381594   32.204208  
std      0.806057   49.693429  
min      0.000000    0.000000  
25%      0.000000    7.910400  
50%      0.000000   14.454200  
75%      0.000000   31.000000  
max      6.000000  512.329200  

Survival Count
Survived
0    549
1    342
Name: count, dtype: int64

Passenger Class
Pclass
3    491
1    216
2    184
Name: count, dtype: int64

Average Age: 29.69911764705882
```